Gautam Lodha

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GitHub | LinkedIn | CodeChef | LeetCode | Codeforces

PROFESSIONAL SUMMARY

CS undergraduate focused on backend and systems engineering, with hands-on experience building message queues, rate limiters, and containerized code execution pipelines through personal projects. Comfortable across the **SDLC**, with practical experience in **Node.js**, **TypeScript**, **Docker**, and **AWS**. Strong foundation in **DSA**, **OOPs**, **DBMS**, and **Operating Systems** – **LeetCode Knight** (peak 1903). Seeking Systems Engineer / SWE / Backend roles.

EDUCATION

LNCT | Bhopal Aug 2023 – Present
B.Tech in Computer Science Engineering

Modern Children School | Guna
Higher Secondary (Class XII) – **93%**

PROJECTS

Real-Time Quiz & Leaderboard Platform *Live*
React, TypeScript, Node.js, Express, Socket.IO, PostgreSQL, Redis, Prisma, BullMQ, JWT
- Built a real-time quiz app with room codes and a **dynamic leaderboard** (correctness + response time) via **Socket.IO**; JWT-authenticated with quiz scheduling through **BullMQ + Redis**.
- Designed a normalized **PostgreSQL** schema via **Prisma ORM**; deployed frontend on **Vercel** and backend on **Render** independently.

Neural IDE – Online Multi-Language Code Compiler *Demo*
Node.js, Bun, TypeScript, Express.js, BullMQ, Redis, Docker, Dockerode
- Built an online code execution engine for **C++, Java, Python, and JavaScript** using a producer-consumer architecture with a **BullMQ + Redis** queue to decouple the API from execution.
- Engineered **sandboxed Docker execution** with CPU/memory/PID limits, non-root users, disabled networking, and a read-only filesystem to prevent fork bombs and sandbox escape.

Distributed API Rate Limiter & Gateway *Link*
Node.js, TypeScript, Redis, Express.js
- Engineered a distributed **Token Bucket** rate limiter with **Redis** for synchronized, thread-safe request control across clients in a standalone API gateway.
- Implemented timestamp-based token refill to guarantee fair distribution and prevent starvation under bursty traffic.

SKILLS

Languages:	C++, Java, JavaScript (ES6+), TypeScript, SQL
Backend:	Node.js, Express.js, REST APIs, Microservices, Socket.IO, BullMQ, Message Queues
Databases & Caching:	PostgreSQL, MongoDB, Redis, Prisma ORM
Frontend:	React.js, Next.js, Tailwind CSS
Cloud & DevOps:	AWS, CI/CD, Docker, Git, GitHub, Postman, Vercel
Tools & Practices:	Agile/Scrum, SDLC
CS Fundamentals:	DSA, OOPs, DBMS, Operating Systems
Soft Skills:	Problem Solving, Analytical Thinking, Communication, Team Collaboration

AWARDS & ACHIEVEMENTS

- **LeetCode Knight** – Peak rating **1903**
- **Codeforces Pupil** – Peak rating **1235**
- **CodeChef 2⋆** – Rating **1543**
- Ranked #**370** in CodeChef Starters 236 (Div. 3)
- **YIDS Hackathon Finalist**